

D.A.P. System Test Procedure

29MAY2026

Document Information

Document Title	D.A.P. System Test Procedure
Document ID	DAP-001
Version	1.4
Author(s)	DW, LL, HK, JC
Approver(s)	
Date	29MAY26

Test Team Personnel:

Position Name	Field or Pilot Team	Name
Pilot	Pilot Team	Claire Syrett

Contents

1	Test Objective	3
2	Test Procedure	4
2.1	Equipment	4
2.2	Phase B Test Setup	4
2.3	Phase B Procedure	5
2.3.1	DAP System Post-Processing Procedure	5
2.4	Shutdown	9
3	Acceptance Criteria	9

1 Test Objective

The objective of this Phase B test procedure is to evaluate the performance of the Damage Assessment Protocol (D.A.P.) system using the mission `.nc` and `.log` files defined in the Test Matrix. Because the planned Phase A field deployment did not yield usable dive data due to Seaglider technical issues, all system-level evaluation is conducted through post-processing of previously recorded mission datasets. Phase B consists solely of running each Test Matrix case through the calibrated Digital Twin and Diagnoser to determine whether the system correctly identifies nominal, wing-loss, or rudder-loss physical faults.

During this phase, the Digital Twin is initialized using the recorded initial conditions from each `.nc` and `.log` file and is used to generate predicted vehicle behavior for the full mission. The Diagnoser compares these predictions to the mission data to compute percent-difference metrics, window scores, excursion intervals, and a final classification. Runtime is recorded for each test case to evaluate computational performance.

The system requirements verified in this phase are:

- **SR-1:** The system shall only utilize onboard Seaglider sensor data, demonstrated by requiring exclusively the mission `.nc` and `.log` files as inputs for all Digital Twin and Diagnoser processing.
- **SR-2:** The system shall diagnose single wing-loss and rudder-loss physical faults within five data transmissions of fault detection.
- **SR-4:** The system shall process a single dive dataset in under one hour of computational runtime.

The outputs generated during Phase B include classification results and runtime measurements. These outputs are archived and used to validate compliance with SR-1, SR-2, and SR-4.

2 Test Procedure

2.1 Equipment

Table 1: Equipment Required for D.A.P. System Test

Item	Description	Value
Seaglider Piloting Laptop or workstation with MATLAB	Must have MATLAB R2024b or later installed; must be able to run .m scripts and load .eng files; no calibration required	1
coefficientsV4_25wing.m	flight coefficients for 25% wing seaglider	1
coefficientsV4_50wing.m	flight coefficients for 50% wing seaglider	1
coefficientsV4_50rudder.m	flight coefficients for 50% rudder seaglider	1
coefficientsV4_nominal.m	flight coefficients for nominal case seaglider	1
Create_Mission_Data_Matrix.m	Compiles relevant mission data into matrix form for analysis	1
D_fault_isolator.m	Isolates fault based on data analysis	1
DAP_Main_Full_System_Test.m	Main file to call all functions within the DAP system	1
Digital_Twin.m	Calls the dynamics code to simulated seaglider states based on fault cases. compiles data into matrix form	1
dynamicsV7.m	Code to simulate various kinematic states of the seaglider	1
Log_File_Unpacker.m	Unpacks relevant data from log files	1
missioncompare_detector.m	Compares mission data to simulated data to determine if a error bound of 40% is prevalent	1
NC_File_Unpacker.m	Unpacks relevant data from NC files	1
parametersV5.m	Set parameters used by dynamics code	1
Log_and_NC_Files	This directory includes the Colvos_Dives, Edmonds_Dives, and Motive_Dives folders, which contain all mission-specific .log and .nc files required for the test.	1

2.2 Phase B Test Setup

Before beginning the Phase B post-processing evaluation, the following setup steps must be completed:

1. ____ Turn on Seaglider Piloting Laptop.
2. ____ Open MATLAB.
3. ____ Confirm that the MATLAB version by typing 'ver' into the command window.

Is MATLAB version R2024b? ____

4. ____ Confirm that the "D.A.P. Test Version" file has the complete list of functions:

coefficientsV4_25wing.m	OK? ____
coefficientsV4_50wing.m	OK? ____
coefficientsV4_50rudder.m	OK? ____
coefficientsV4_nominal.m	OK? ____
Create_Mission_Data_Matrix.m	OK? ____
D_fault_isolator.m	OK? ____
DAP_Main_Full_System_Test.m	OK? ____
Digital_Twin.m	OK? ____
dynamicsV6_06.m	OK? ____
Log_File_Unpacker.m	OK? ____
missioncompare_detector.m	OK? ____
NC_File_Unpacker.m	OK? ____
parametersV5.m	OK? ____
Log_and_NC_Files directory	OK? ____

5. ____ Confirm that the Colvos_Dives, Edmonds_Dives, and Motive_Dives folders are present in the Log_and_NC_Files directory within the DAP_Test_Version.

All files present? ____

6. ____ Navigate to DAP_Main_Full_System_Test.m file and open it in MATLAB

2.3 Phase B Procedure

2.3.1 DAP System Post-Processing Procedure

1. ____ Identify the fault deviation gain, the dives folder name and the .nc and .log file name from the Test Matrix.

Fault deviation gain confirmed? ____

Dives folder name confirmed? ____

.nc and .log file name confirmed? ____

2. ____ Press the RUN button and follow the input prompts.

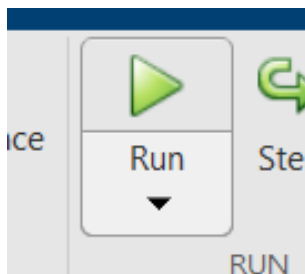


Figure 1: Run Button for Matlab

- (a) When prompted for the .log files, click OK.

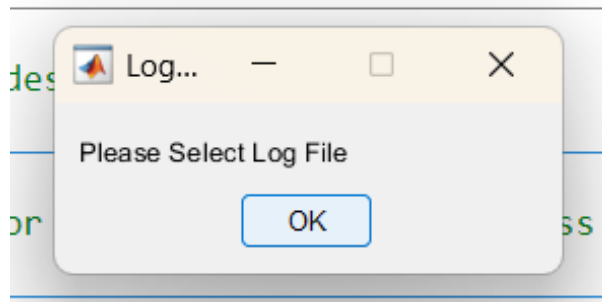


Figure 2: .log file Input Display

- (b) Open the Log_and_NC.Files folder

Name	Date modified	Type	Size
Log_and_NC.Files	5/27/2026 5:07 PM	File folder	
MS_Comparator_Plots	5/27/2026 5:02 PM	File folder	

Figure 3: Example of Log_and_NC.Files Folder Selection

- (c) Open the corresponding Dives folder for the test case.

Name	Date modified	Type
Colvos_Dives	5/27/2026 5:07 PM	File folder
Edmonds_Dives	5/27/2026 5:07 PM	File folder
Motive_Dives	5/27/2026 5:07 PM	File folder

Figure 4: Example of Dives Folder Selection

(d) Select your corresponding .log file

Name	Date modified	Type
p1940003	5/27/2026 3:37 PM	Text Document
p1940004	5/27/2026 3:37 PM	Text Document
p1940005	5/27/2026 3:37 PM	Text Document
p1940006	5/27/2026 3:37 PM	Text Document
p1940007	5/27/2026 3:37 PM	Text Document

Figure 5: Example of .log File Selection

(e) When prompted for .nc file, click OK.

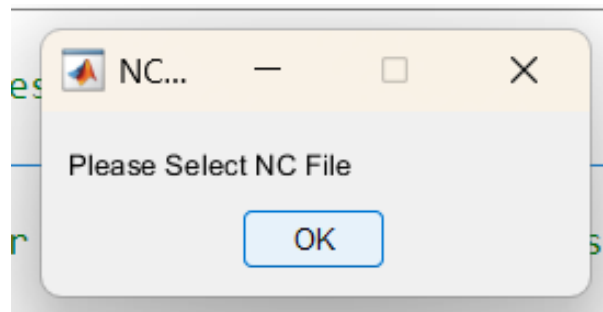


Figure 6: .nc file Input Display

(f) Open the Log_and_NC_Files folder

(g) Open the corresponding Dives folder for the test case.

(h) Select your corresponding .nc file

(i) When prompted, enter the Fault Deviation Gain value (0, 1, or 2).

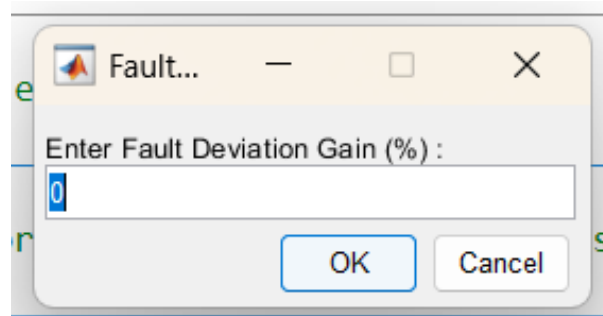


Figure 7: Fault Deviation Gain Input Display

3. ____ The D.A.P Code will run once the test case is inputted. Wait until final damage assessment report appears in the command window.

Code executing? ____

Final Damage Assessment Report Generated? ____

4. ____ Record the Primary Diagnosis in the Test Matrix

Primary Diagnosis recorded? ____

```
=====
FINAL DAMAGE ASSESSMENT REPORT
=====
Primary Diagnosis: 50% Wing Fault
Confidence Metric (Error): 42.68%
      Fault_Case      Average_Percent_Error
      -----      -
{'50% Wing Fault'  }      42.678
{'25% Wing Fault'  }      42.678
{'50% Rudder Fault'}      42.678
=====
Runtime: 143.91 seconds
```

Figure 8: Example of where to find Primary Diagnosis

5. ____ Record the Runtime in the Test Matrix

Runtime recorded? ____

```
=====
FINAL DAMAGE ASSESSMENT REPORT
=====
Primary Diagnosis: 50% Wing Fault
Confidence Metric (Error): 42.68%
      Fault_Case      Average_Percent_Error
      -----      -
{'50% Wing Fault'  }      42.678
{'25% Wing Fault'  }      42.678
{'50% Rudder Fault'}      42.678
=====
Runtime: 143.91 seconds
```

Figure 9: Example of where to find the Runtime

6. ____ Repeat Steps 1–5 for all remaining test cases in the Test Matrix.

2.4 Shutdown

1. ____ Once all test cases have been processed, archive all output files and close MATLAB R2024b.

Files archived? ____

MATLAB closed? ____

2. ____ Shutdown Piloting Laptop

3 Acceptance Criteria

The following Yes/No acceptance criteria are recorded for each Phase B test run:

1. (Yes/No) Primary Diagnosis (Nominal, Wing Fault, or Rudder Fault) was recorded.
2. (Yes/No) The total runtime required to process the .nc file was successfully recorded.

Version History

Version	Date	Author	Description of Changes
1.0	25MAY26	DW	Initial Draft
1.1	26MAY26	JC	Added equipment list
1.2	27MAY26	JC	Adjusted procedures based on code script
1.3	27MAY26	DW, LL, JC	Adjusted procedure to include pictures of how to run the system
1.4	28MAY26	LL	Combined verifying MATLAB version in the Test Setup into one step, resized pictures